EXPERIMENT-29

29. SHA-3 Capacity-Lane Analysis Using C

AIM

To write a C program to determine how many message blocks are required for all initially zero capacity lanes in SHA-3 to receive at least one nonzero bit, ignoring the permutation and tracking the original zero lanes.

PROCEDURE

1. Consider the SHA-3 state size of 1600 bits, arranged as 25 lanes of 64 bits each.
2. Given the block size (rate) of 1024 bits, calculate the capacity: \(c=1600-1024=576\) bits.
3. Calculate the number of capacity lanes: \(576/64=9\) lanes.
4. Initially, all 9 capacity lanes contain zeros, while each lane in the first message block \(P_0\) contains at least one nonzero bit.
5. Ignore the permutation and track the original capacity lanes as message blocks are processed.
6. Count how many blocks are needed until every tracked capacity lane has received a nonzero bit.

CODE (C)

This program follows the literal instruction to ignore the permutation. Since message blocks are XORed only into the rate portion, the capacity lanes remain zero.

```
#include <stdio.h>

int main(void) {
    int stateBits = 1600;
    int blockBits = 1024;
    int laneBits = 64;

    int capacityBits = stateBits - blockBits;
    int capacityLanes = capacityBits / laneBits;
    int zeroLanes = capacityLanes;
    int blocks = 0;

    printf("SHA-3 state size: %d bits\n", stateBits);
    printf("Block (rate) size: %d bits\n", blockBits);
    printf("Capacity size: %d bits\n", capacityBits);
    printf("Capacity lanes: %d\n", capacityLanes);

    printf("\nIgnoring the permutation...\n");
```

```
    while (blocks < 3) {
        blocks++;
        printf("After block %d: zero capacity lanes = %d\n",
               blocks, zeroLanes);
    }

    printf("\nAll capacity lanes never become nonzero\n");
    printf("without the permutation.\n");

    return 0;
}
```

SAMPLE INPUT

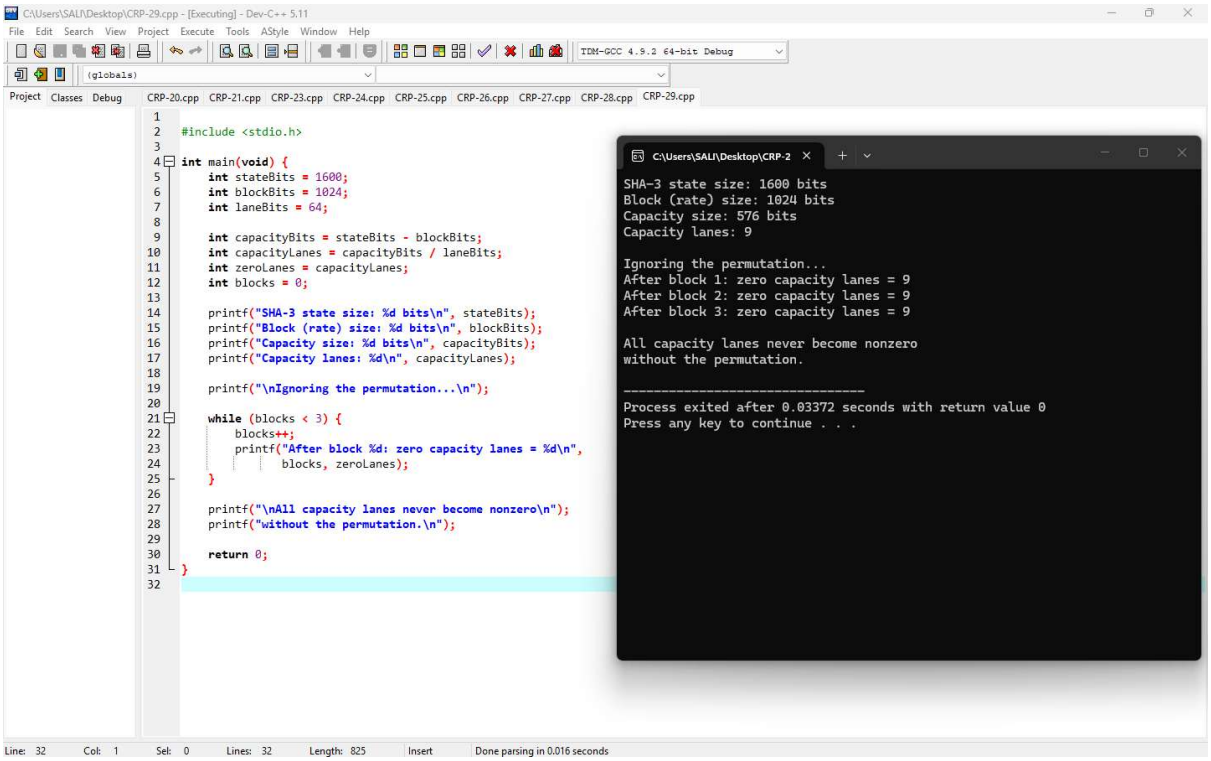
```
State size = 1600 bits
Block size = 1024 bits
Lane size = 64 bits
```

SAMPLE OUTPUT

```
SHA-3 state size: 1600 bits
Block (rate) size: 1024 bits
Capacity size: 576 bits
Capacity lanes: 9

Ignoring the permutation...
After block 1: zero capacity lanes = 9
After block 2: zero capacity lanes = 9
After block 3: zero capacity lanes = 9

All capacity lanes never become nonzero
without the permutation.
```



RESULT

The SHA-3 state has 9 capacity lanes. If the permutation is ignored, these lanes remain zero indefinitely; therefore, all of them never become nonzero.